[NiFe]-hydrogenase + H_2 — v6-small pilot run

LAMMPS analysis figures (16)

Important: every H_2 quantity in this run refers to a ~3 Å rigid dumbbell, not H_2 (H–H separation frozen at 2.08–3.32 Å instead of 0.741 Å; see Fig. 8). Protein, water and thermostat results are usable.

Fig. 1: Production overview (250 ps): T(H2), contacts, closest H2 to Ni, MSD, Rg, RMSD proxy
Fig. 2: MSD per axis and log-log MSD (single time origin)
Fig. 3: H2 temperature distribution vs canonical expectation; autocorrelation
Fig. 4: H2-protein contact count, distribution, memory
Fig. 5: Closest H2 to the Ni-Fe centre
Fig. 6: Protein Rg and RMSD proxy vs Kabsch RMSD
Fig. 7: NPT equilibration: density, Rg, RMSD, H2-Ni distances
Fig. 8: Final frame: H2 positions, hydration water density, H-H geometry check (FAILED)
Fig. 9: Final frame projections
Fig. 10: Protein over NPT + production: Kabsch RMSD, Rg, distance to own periodic image
Fig. 11: Per-atom RMSF and RMSF vs distance from Ni
Fig. 12: H2 COM MSD, all time origins; per-molecule D
Fig. 13: Per-molecule distance to Ni and water/contact/buried state
Fig. 14: Contact episodes and residence correlation functions
Fig. 15: RDF H2 → protein atoms and running coordination number
Fig. 16: Cross-check: trajectory analysis vs LAMMPS on-the-fly output

Fig. 1

Production overview (250 ps): T(H2), contacts, closest H2 to Ni, MSD, Rg, RMSD proxy

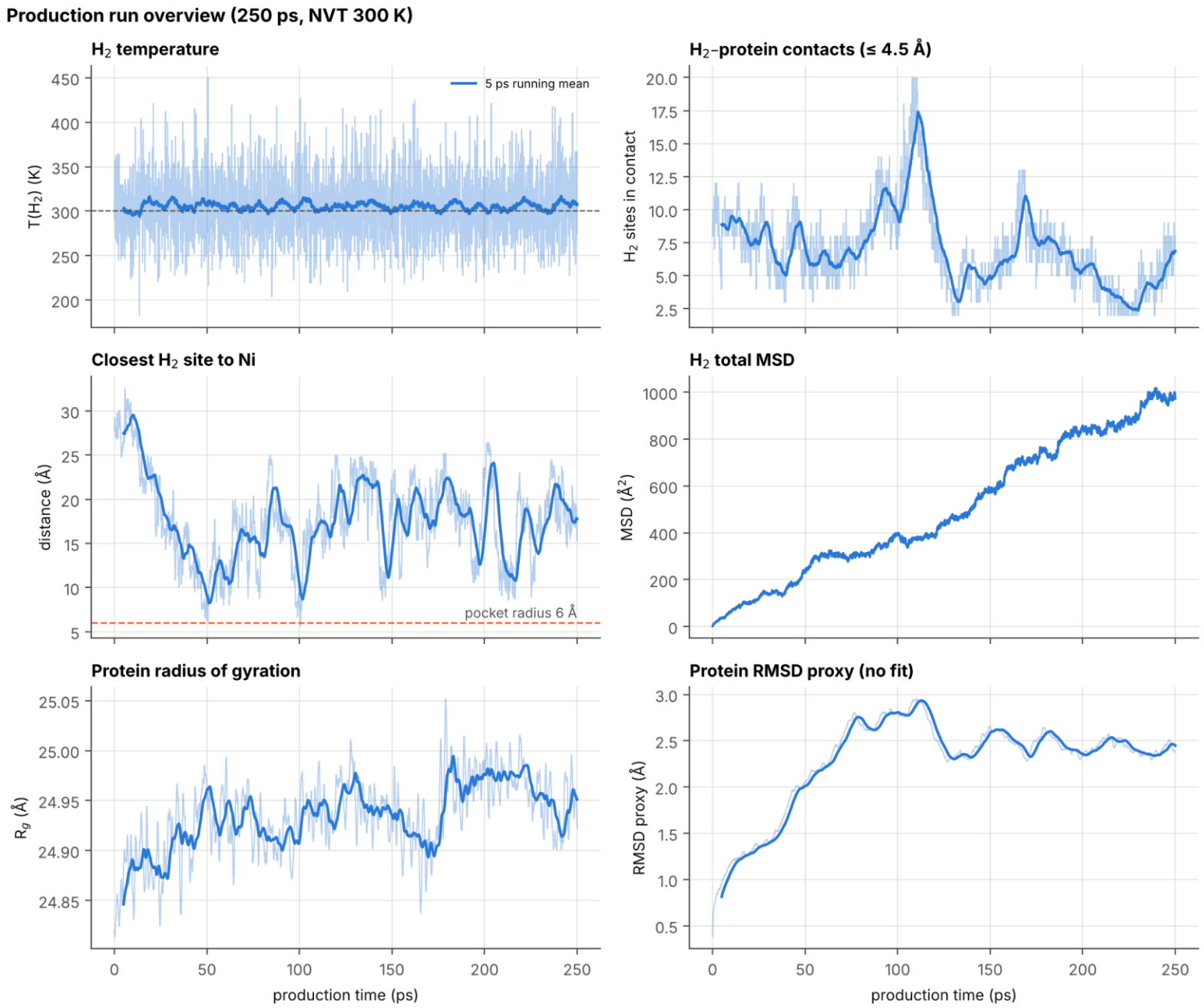


Fig. 2

MSD per axis and log-log MSD (single time origin)

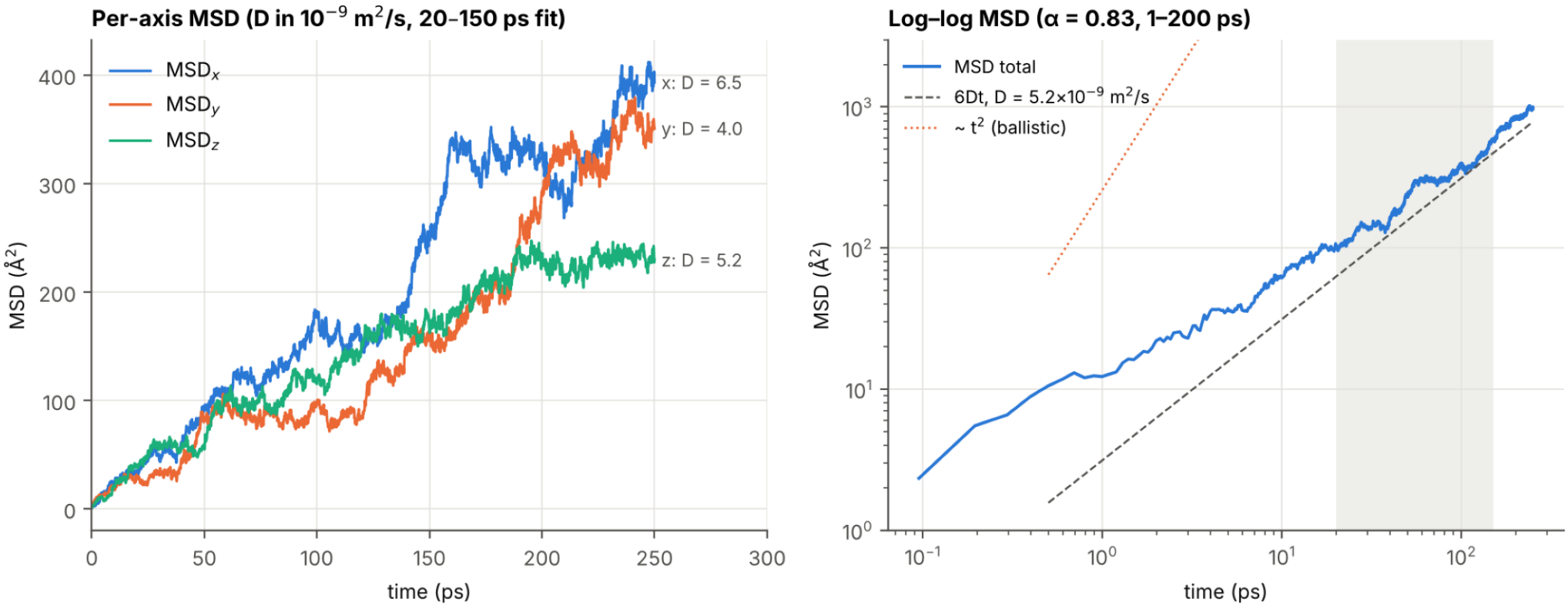


Fig. 3

H2 temperature distribution vs canonical expectation; autocorrelation

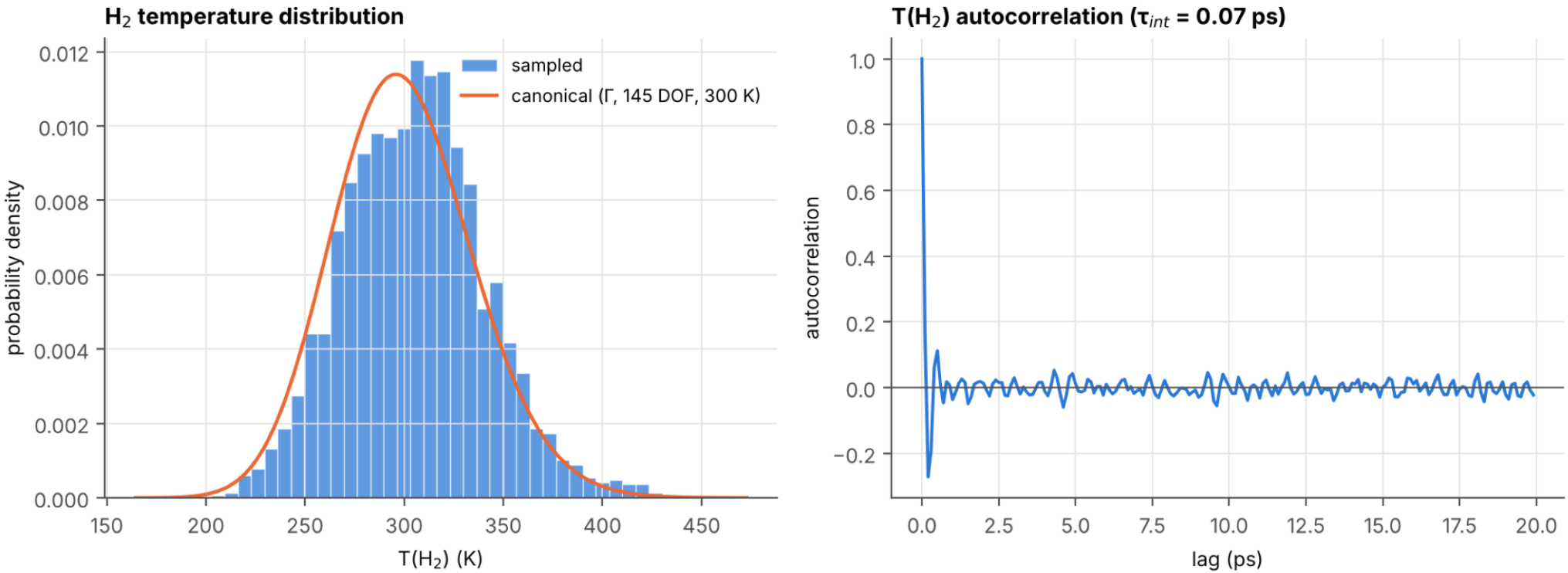


Fig. 4

H2-protein contact count, distribution, memory

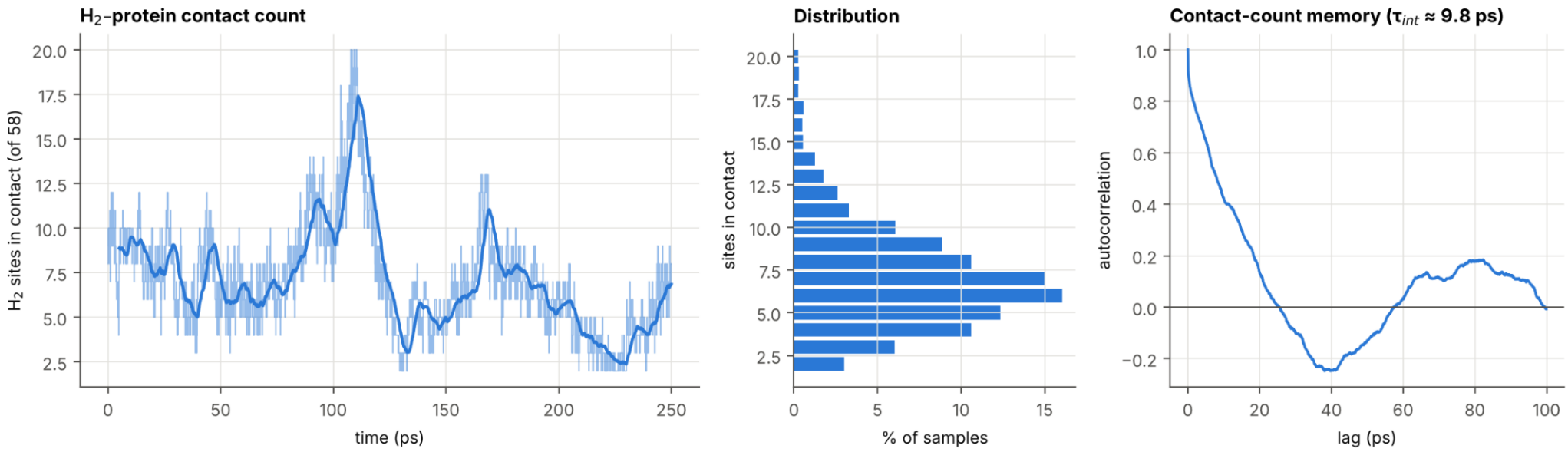


Fig. 5

Closest H2 to the Ni-Fe centre

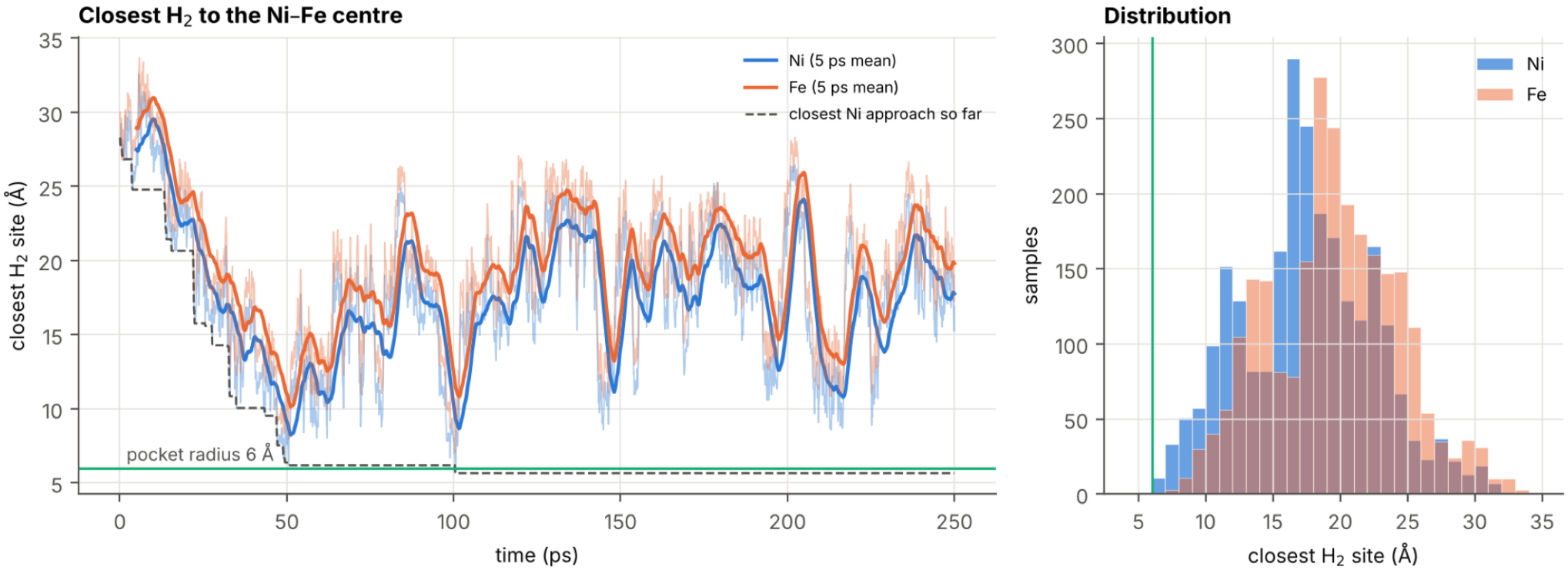


Fig. 6

Protein Rg and RMSD proxy vs Kabsch RMSD

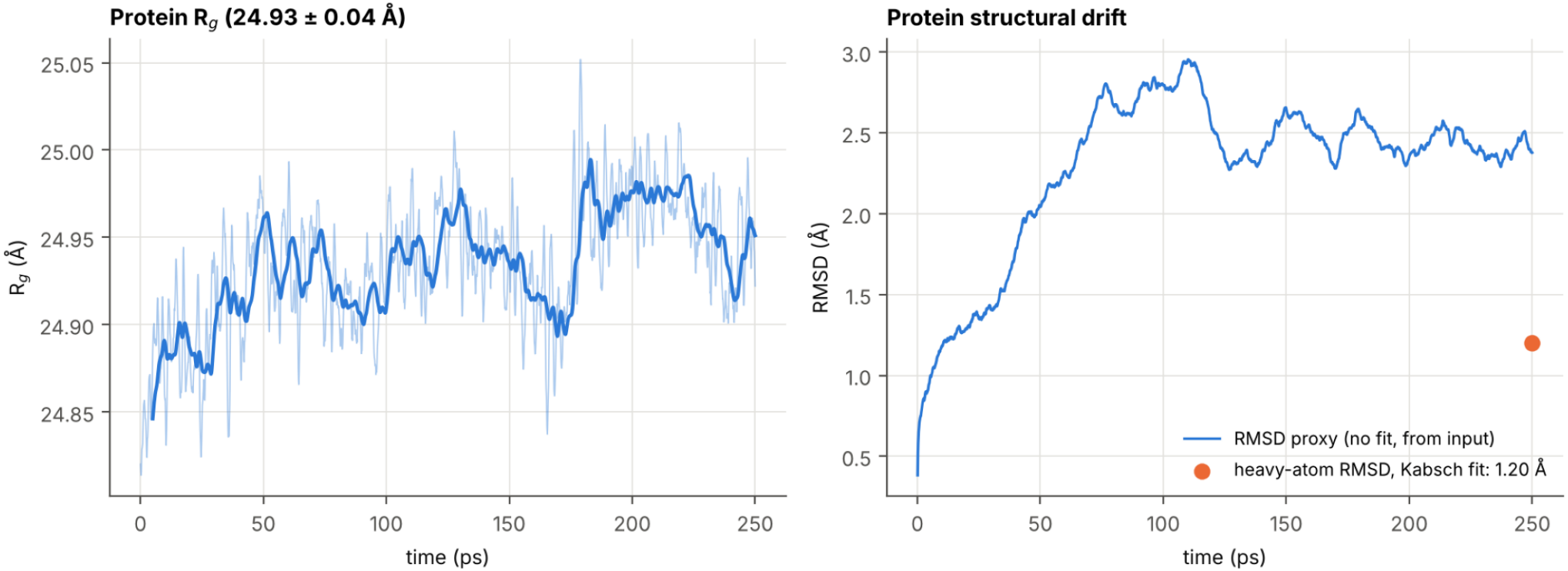


Fig. 7

NPT equilibration: density, Rg, RMSD, H2-Ni distances

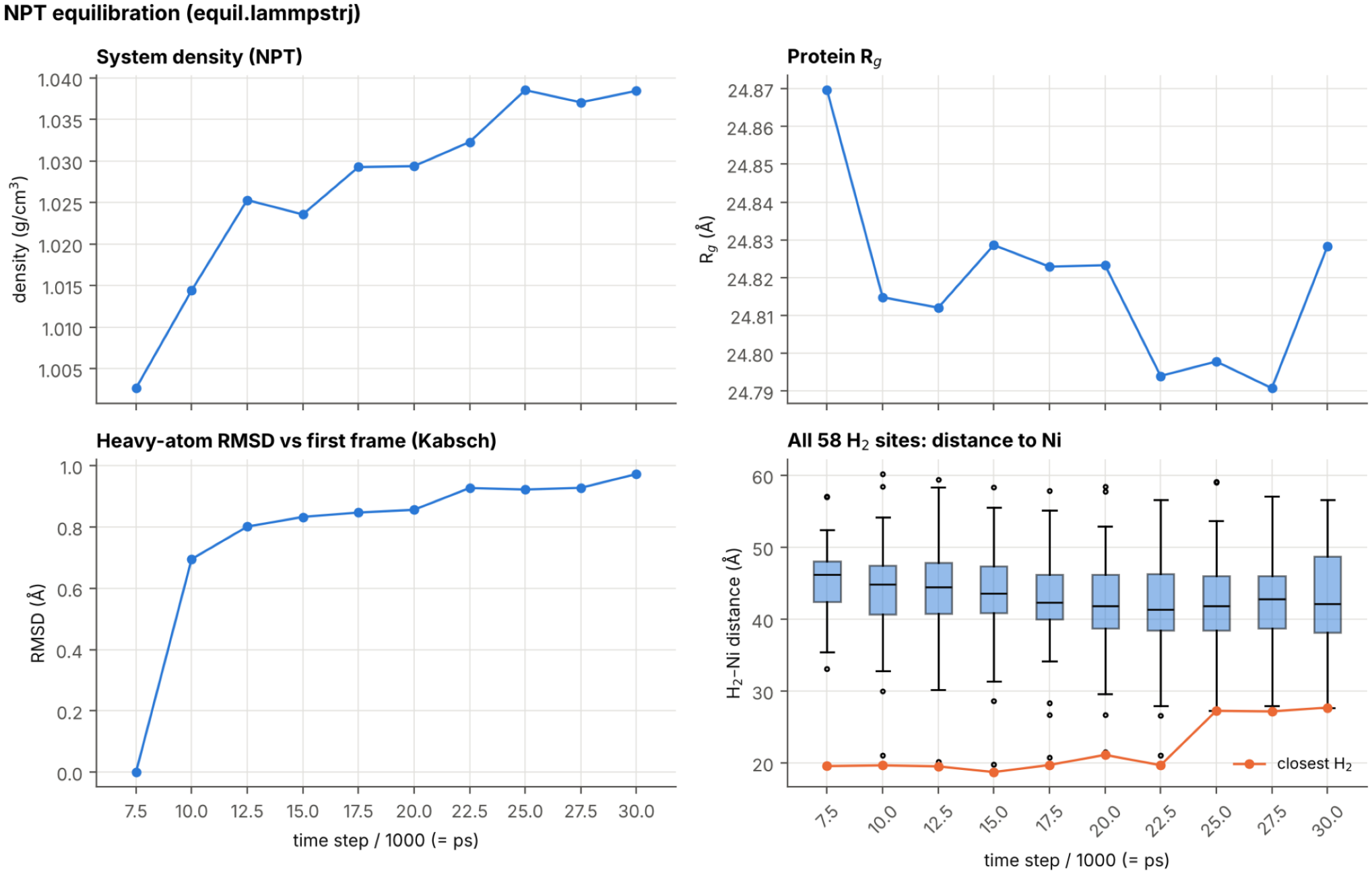


Fig. 8

Final frame: H2 positions, hydration water density, H-H geometry check (FAILED)

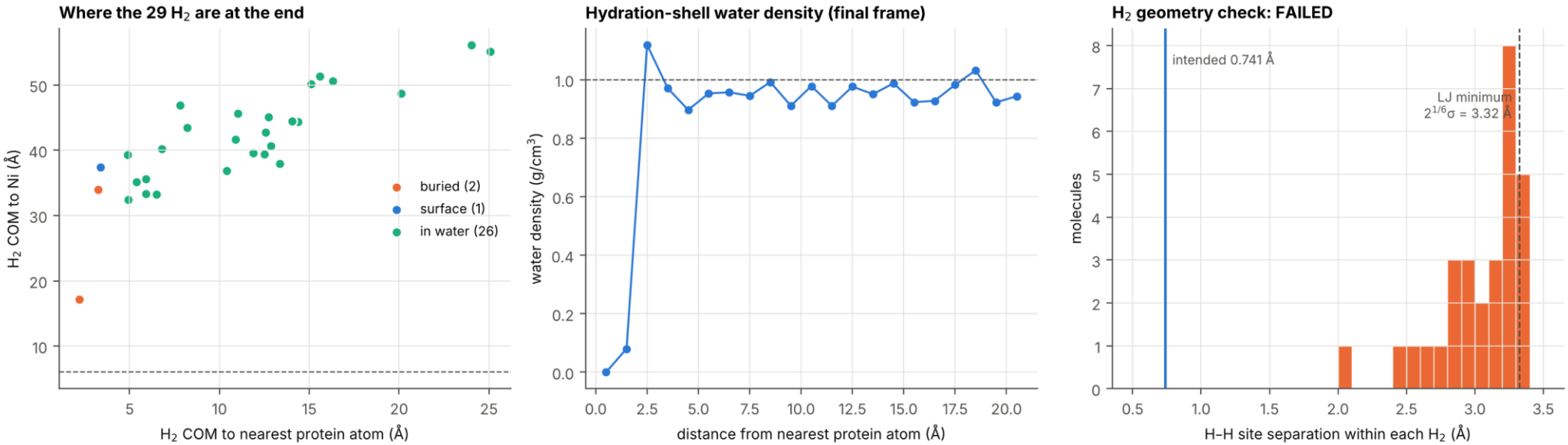


Fig. 9

Final frame projections

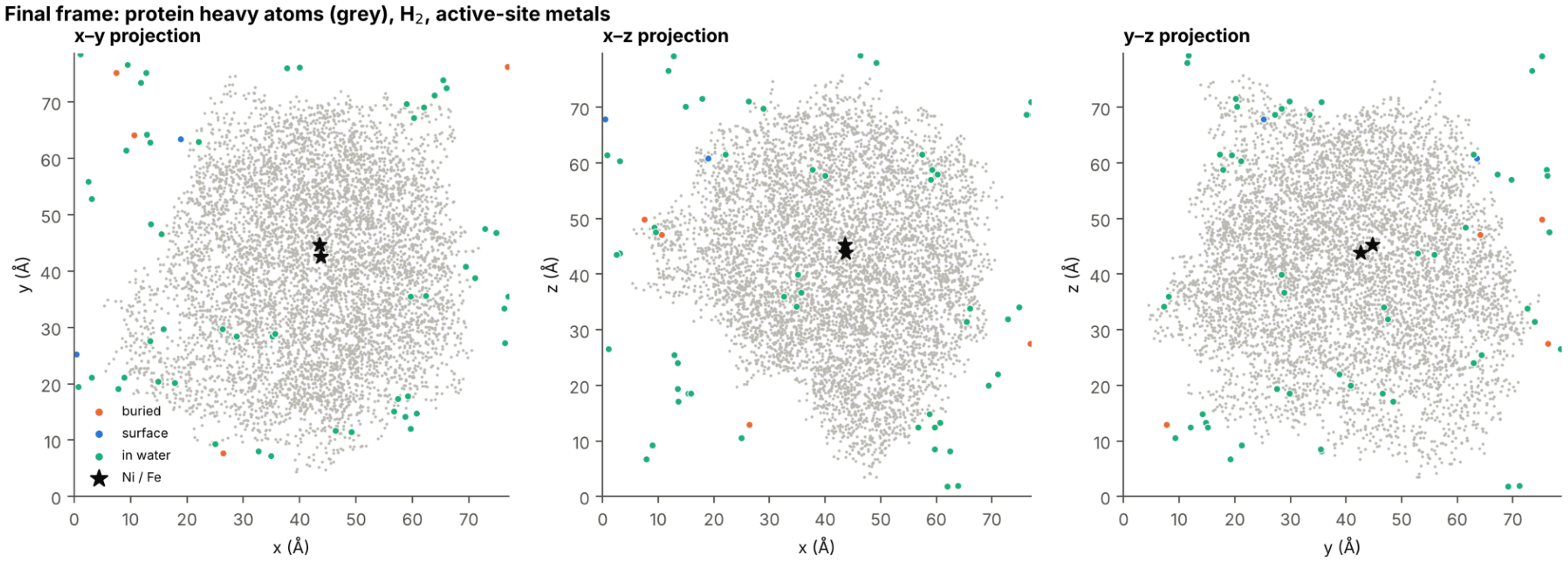


Fig. 10

Protein over NPT + production: Kabsch RMSD, Rg, distance to own periodic image

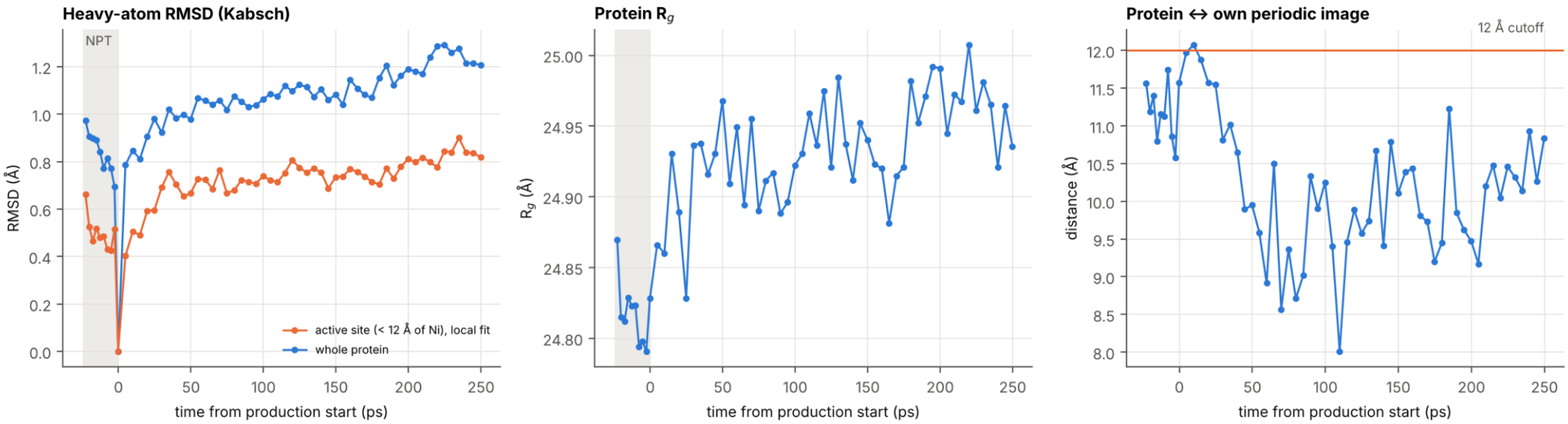


Fig. 11

Per-atom RMSF and RMSF vs distance from Ni

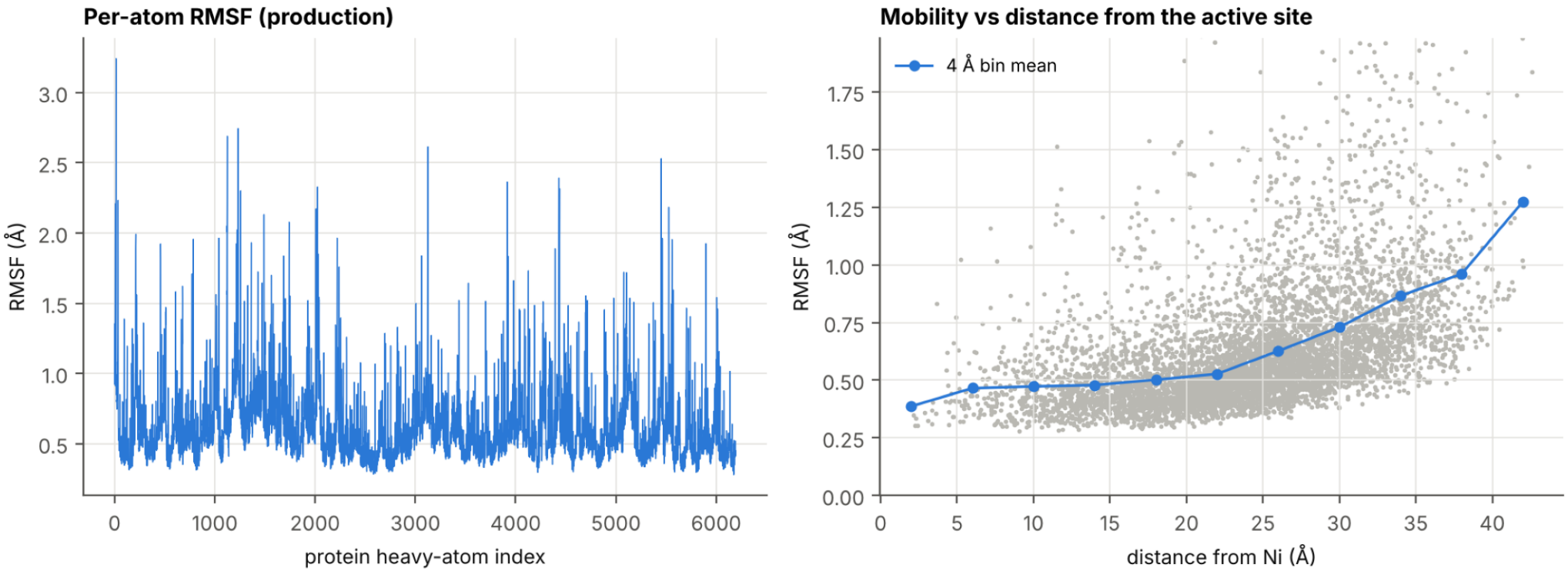


Fig. 12

H2 COM MSD, all time origins; per-molecule D

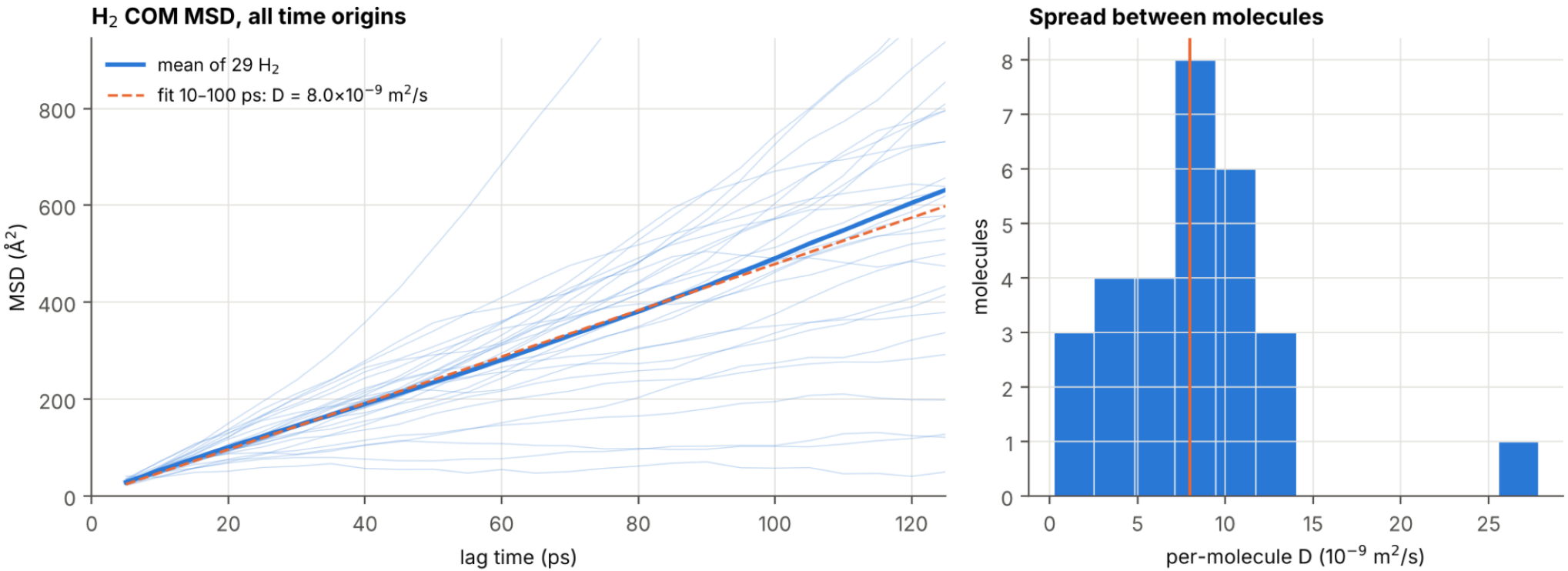


Fig. 13

Per-molecule distance to Ni and water/contact/buried state

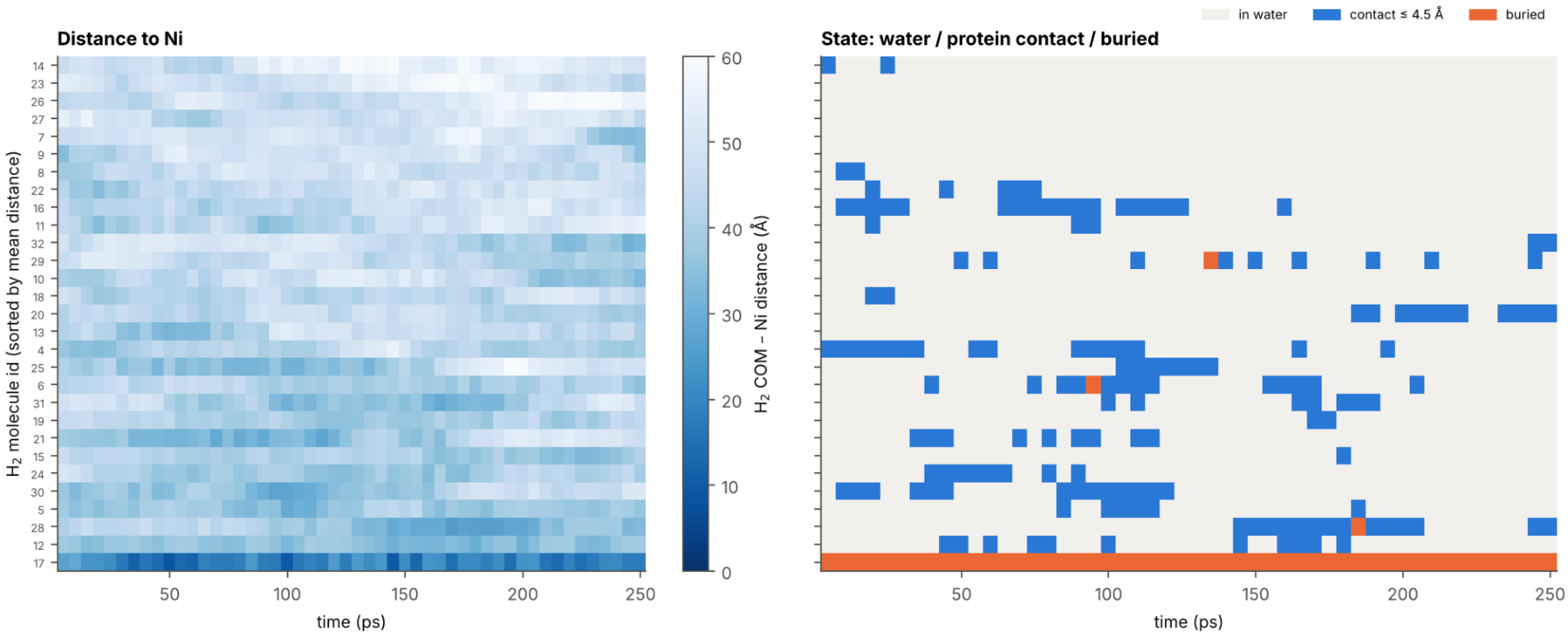


Fig. 14

Contact episodes and residence correlation functions

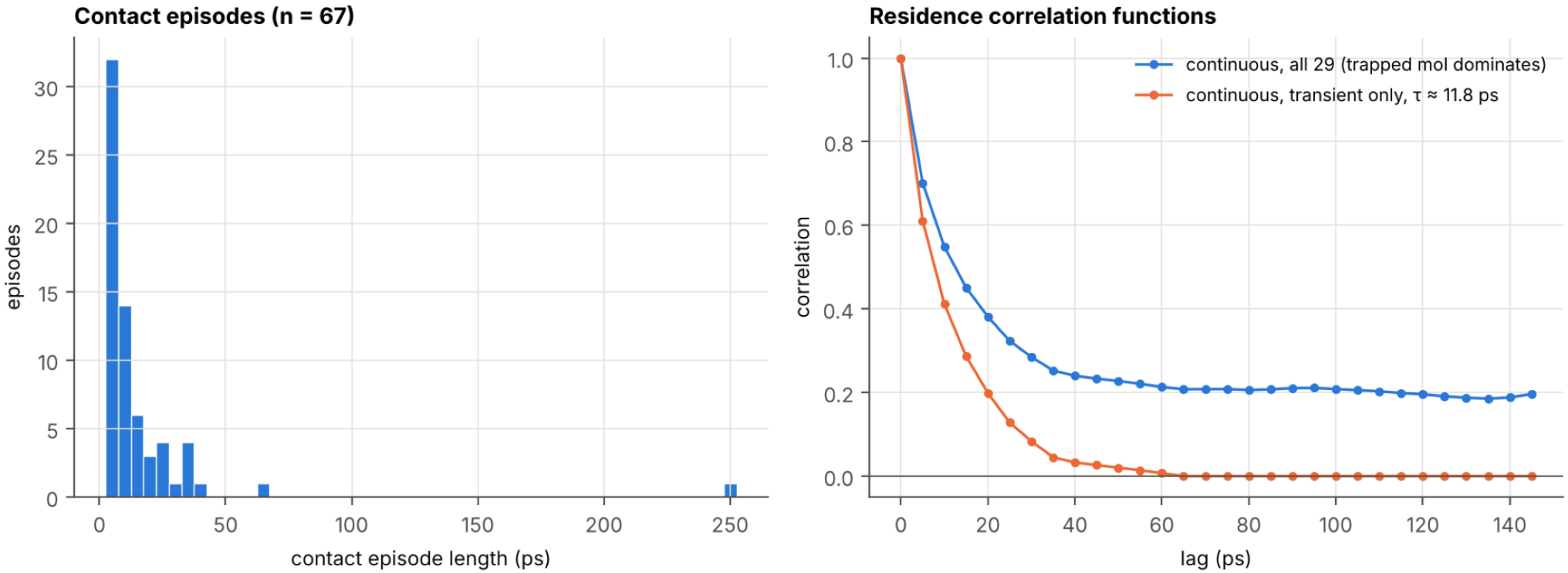


Fig. 15

RDF H2 → protein atoms and running coordination number

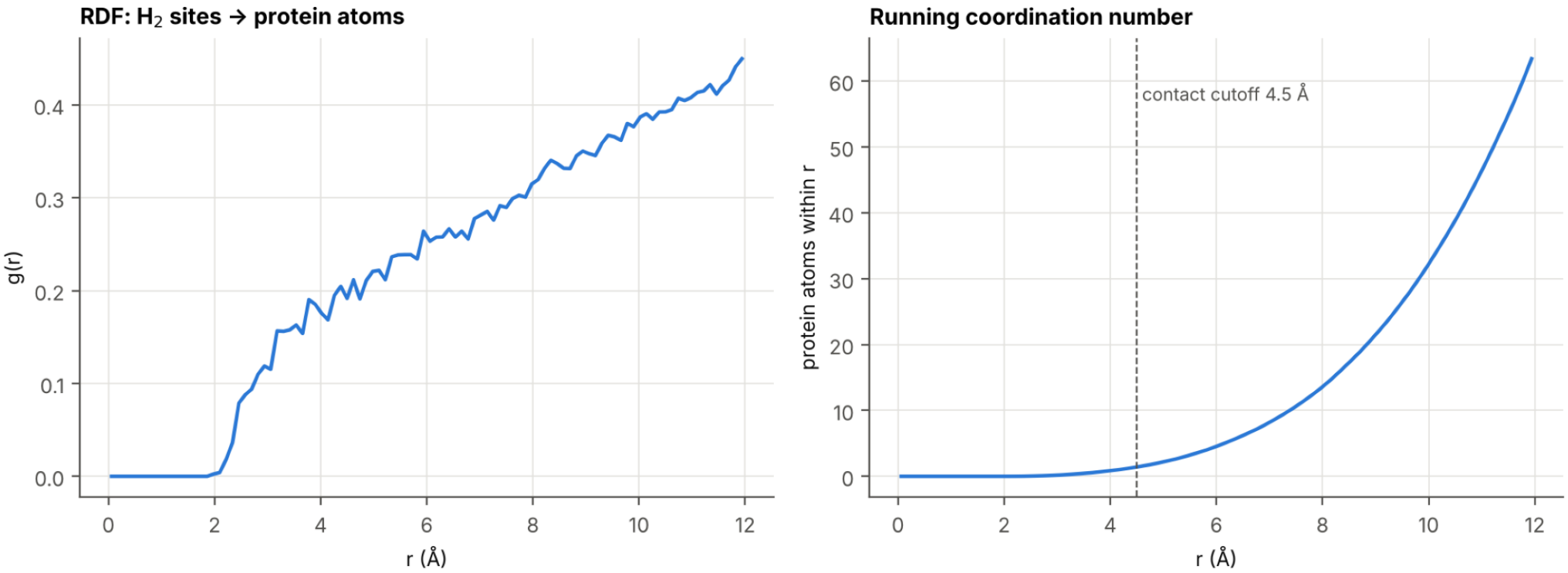


Fig. 16

Cross-check: trajectory analysis vs LAMMPS on-the-fly output

